

DAY-4

NETWORK TABLE AND TYPE AND TOOL THEORY

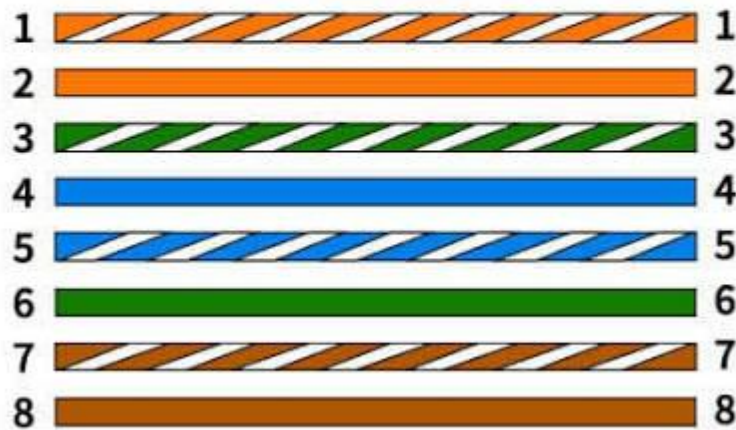
1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

<i>Feature</i>	<i>Ethernet cable</i>	<i>Fiber optic cable</i>
<i>Material and signal</i>	<i>Usually uses copper wires and electrical signals.</i>	<i>Uses glass or plastic fibers and light signals.</i>
<i>Distance</i>	<i>Best for shorter connections, generally</i>	<i>Supports much longer distances, from kilometers to much</i>

<i>Feature</i>	<i>Ethernet cable</i>	<i>Fiber optic cable</i>
	<i>up to about 100 meters.</i>	<i>farther with suitable equipment.</i>
<i>Interference and cost</i>	<i>Cheaper and easier to install, but can be affected by electromagnetic interference.</i>	<i>More expensive and requires specialized equipment, but is resistant to interference and supports high bandwidth. walsun+1</i>
<i>Example use</i>	<i>Connecting a computer or printer to a home Wi-Fi router.</i>	<i>Connecting buildings across a college campus or linking servers in a data center. <u>fiber-life</u></i>

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

Straight Through Cable T568B



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

1. *Cut the CAT6 cable to the required length and place a cable boot on it first, if one is being used.*
2. *Carefully remove about 2–3 cm of the outer jacket. Do not cut deeply, or the small wires inside may be damaged.*
3. *Separate the four twisted pairs and straighten the eight individual wires.*

4. *Arrange the wires in the T568B order: white/orange, orange, white/green, blue, white/blue, green, white/brown, brown.*
5. *Hold the wires tightly in that order and trim their ends so they are even.*
6. *Slide the wires into the RJ45 connector, keeping the connector clip facing away from you. Check that each wire reaches the front of its own channel and that the cable jacket enters far enough inside the plug.*
7. *Insert the connector fully into the correct RJ45 slot on the crimping tool.*
8. *Squeeze the crimper firmly once to push the metal contacts into the wires and secure the cable jacket.*
9. *Inspect the plug to ensure the wires are still in the correct order and the jacket is held by the connector.*
10. *Make the other end using the same T568B order for a straight-through cable, then test the cable with a network cable tester. A correct cable should show pins 1–8 in the proper sequence without opens or crossed wires.*

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45

crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.

For a gaming café with only 10 PCs in a single room or small space, I would choose Ethernet (Cat6 or Cat6a copper) cables and an RJ45 crimper.

Distances are short (well under 100 meters), so copper Ethernet easily delivers the 1 Gbps or multi-Gigabit speeds needed for low-latency LAN gaming at a much lower cost than fiber. Fiber would be overkill and significantly more expensive due to the need for optical transceivers, specialized connectors, and more complex installation. An RJ45 crimper lets me quickly make custom-length cables on-site, while a fiber cleaver is unnecessary for this short-range copper setup.